

Code-Layout, Readability and Reusability

Date	7 JULY 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper indentation is maintained throughout the Python and Flask code for better readability.
2	Proper File Structure	Files and folders are logically organized	yes	The project is organized into folders such as templates, static, dataset, model, and application files.
3	Meaningful Variable Name	Variables reflect their purpose clearly	yes	Variables like rainfall, cloud_visibility, seasonal_rainfall, and prediction_result clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	yes	Functions such as load_model(), predict_flood(), and preprocess_data() use descriptive names.
5	Code Comments	Inline and block comments explain logic	yes	Important sections include comments explaining preprocessing, model prediction, and application logic.
6	Modular Design	Code is split into reusable functions/modules	yes	Data preprocessing, model prediction, and web interface are implemented as separate reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	yes	Duplicate code has been minimized by using reusable functions and methods.
8	Error Handling	Exceptions and errors are handled gracefully	Cefully yes	Input validation and exception handling are implemented to prevent application crashes.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Preparing input data before prediction	High
2	Flood Prediction Model	Scikit-learn, XGBoost	Predicting flood risk from weather data	High
3	Flask Web Application	Flask (Python)	User interface and prediction service	High
4	Model Loading Module	Pickle, Python	Loading the trained ML model for predictions	Medium
5	Prediction Result Display	HTML, CSS, Flask	Displaying prediction results to users	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5/5	The project follows a well-organized folder structure with separate modules for data, models, and web application files.
Readability	5/5	Meaningful variable names, proper indentation, and comments make the code easy to understand.
Reusability	5/5	Machine learning, preprocessing, and prediction modules can be reused in future projects.
Documentation / Comments	4/5	Code includes comments and documentation for important functions and modules.
Overall Score	4.8 / 5	The project follows good coding practices, making it maintainable, reusable, and easy to extend.